10. A tank initially contains 50 litres of pure water. Starting at time $t = 0$ a brine containing with 2 grams of dissolved salt per litre flows into the tank at the rate of 3 litres per minute. The mixture is kept uniform by stirring and the well-stirred mixture simultaneously flows out of the tank at the same rate. Find the amount of salt present in the tank at any time $t > 0$.

EXERCISE 10.9

B3K7M7

Choose the correct or the most suitable answer from the given four alternatives :

1. The order and degree of the differential equation $\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^{1/3} + x^{1/4} = 0$ are respectively

   (1) 2, 3 (2) 3, 3 (3) 2, 6 (4) 2, 4

2. The differential equation representing the family of curves $y = A\cos(x + B)$, where A and B are parameters, is

   (1) $\frac{d^2y}{dx^2} - y = 0$ (2) $\frac{d^2y}{dx^2} + y = 0$ (3) $\frac{d^2y}{dx^2} = 0$ (4) $\frac{d^2x}{dy^2} = 0$

3. The order and degree of the differential equation $\sqrt{\sin x}\,(dx + dy) = \sqrt{\cos x}\,(dx - dy)$ is

   (1) 1, 2 (2) 2, 2 (3) 1, 1 (4) 2, 1

4. The order of the differential equation of all circles with centre at (h, k) and radius 'a' is

   (1) 2 (2) 3 (3) 4 (4) 1

5. The differential equation of the family of curves $y = Ae^x + Be^{-x}$, where A and B are arbitrary constants is

   (1) $\frac{d^2y}{dx^2} + y = 0$ (2) $\frac{d^2y}{dx^2} - y = 0$ (3) $\frac{dy}{dx} + y = 0$ (4) $\frac{dy}{dx} - y = 0$

6. The general solution of the differential equation $\frac{dy}{dx} = \frac{y}{x}$ is

   (1) $xy = k$ (2) $y = k\log x$ (3) $y = kx$ (4) $\log y = kx$

7. The solution of the differential equation $2x\frac{dy}{dx} - y = 3$ represents

   (1) straight lines (2) circles (3) parabola (4) ellipse

8. The solution of $\frac{dy}{dx} + p(x)y = 0$ is

   (1) $y = ce^{\int pdx}$ (2) $y = ce^{-\int pdx}$ (3) $x = ce^{-\int pdy}$ (4) $x = ce^{\int pdy}$

9. The integrating factor of the differential equation $\frac{dy}{dx} + y = \frac{1+y}{\lambda}$ is

   (1) $\frac{x}{e^{\lambda}}$ (2) $\frac{e^{\lambda}}{x}$ (3) λe^x (4) e^x

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